

Munjung Kim

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Education & Training	University of Virginia <i>Doctor of Philosophy, Data Science</i> • Advisor: Dr. Yong-Yeol Ahn Aug 2025 - Present
	Indiana University Bloomington <i>Doctor of Philosophy program, Informatics (transferred with advisor)</i> • Advisor: Dr. Yong-Yeol Ahn Aug 2022 - Aug 2025
	Pohang University of Science and Technology <i>Bachelor of Science, Physics</i> Feb 2017 - Feb 2022
Work Experience	Nokia Bell Labs Research Intern <i>Cambridge, UK</i> June 2024– Aug 2024 • Identifying job tasks expected to be impacted by disruptive AI technologies using LLMs, text embeddings, and patent citation network analysis [W2]
	SK Hynix Quality Control Intern <i>Icheon, South Korea</i> Jan 2020– Feb 2020
Research Interest	Machine Learning (graph representation learning, language models) Computational Social Science (Innovation, Collective intelligence, Science of science, AI-agent behavior)
Working Papers	[W1] Munjung Kim and Yong-Yeol Ahn. <i>Mitigating Imitative Falsehoods in LLMs Through Crowdsourcing Techniques</i> . [W2] Munjung Kim , Marios Constantinides, Sanja Šćepanović, Yong-Yeol Ahn, and Daniele Quercia. <i>Augmenting Minds or Automating Routines? The Potential Impact of Disruptive AI Innovations on U.S. Occupations</i> .
Publications	[A1] Munjung Kim , Sadamori Kojaku, and Yong-Yeol Ahn. “Uncovering simultaneous breakthroughs with a robust measure of disruptiveness”. <i>Science Advances</i> 12 (2026). [code] [pip] [blog], eadx3420. doi: 10.1126/sciadv.adx3420 . [A2] Rachith Aiyappa, Xin Wang, Munjung Kim , Ozgur Seckin, Yong-Yeol Ahn, and Sadamori Kojaku. “Implicit Degree Bias in the Link Prediction Task”. <i>Proceedings of the 42nd International Conference on Machine Learning (ICML)</i> . Honolulu, Hawaii, 2025. [A3] Munjung Kim , Tuğrulcan Elmas, and Filippo Menczer. “Toxic Synergy Between Hate Speech and Fake News Exposure”. <i>Proceedings of the 18th International AAAI Conference on Web and Social Media (ICWSM 2024)</i> (2024). [A4] Matthew R DeVerna, Rachith Aiyappa, Wanying Zhao, Manita Pote, Bao Tran Truong, David Axelrod, Aria Pessianzadeh, Zoher Kachwala, Munjung Kim , Ozgur Can Seckin, et al. “A multi-platform collection of social media posts about the 2022 US midterm elections”. <i>Proceedings of the international AAAI conference on web and social media</i> (2023). [A5] Munjung Kim , Jisung Yoon, Woo-Sung Jung, and Hyunuk Kim. “Quantifying the topic disparity of scientific articles”. <i>Companion Proceedings of the Web Conference 2022</i> (2022), pp. 769–773.

Oral Presentations	Invited Talks	
	• KDI School Special Lecture 12/17/2024 • KAIST STP Science of Science Research Seminar 12/18/2024 • SNU Business School Workshop 01/03/2025 • KISTI 01/24/2025 • UVA SDS Advisory Board Meeting 04/17/2026 • <i>Panelist</i>, Understanding and Scoping Future Research on Disruptive Innovation: A Workshop, National Academies of Sciences, Washington, DC 05/20/2026	
	Contributed Talks	
	Munjung Kim , Sadamori Kojaku, and Yong-Yeol Ahn. "A Continuous Embedding Measure of Disruptiveness Reveals Hidden Simultaneous Disruption". IC2S2. 2024.	
	Munjung Kim , Sadamori Kojaku, and Yong-Yeol Ahn. "Quantifying disruptiveness using neural embedding method". Netsci 2023. 2023.	
	Munjung Kim , Jisung Yoon, Hyunuk Kim, and Woo-Sung Jung. "Neural embedding of research papers reveals characteristics of policy research institutes". Networks 2021: A Joint Sunbelt & NetSci Conference. 2021.	
	Munjung Kim . "Quantitative Analysis on Institutions' Activity Policy Research with Embedding Method". Korea Academy of Complexity Studies. 2019.	
	Munjung Kim , Jisung Yoon, and Woo-Sung. "Quantitative Analysis on Institutions' Activity Policy Research with Embedding Method". Korea Academy of Complexity Studies. 2019.	
Honors &Awards	2025 December Graduate Conference , Institute for Humane Studies, 2025/ ca. \$100 Humane Studies Fellowship , Institute for Humane Studies, 2024/ ca. \$5,000	
	Complex Systems Summer School , Santa Fe Institute, 2023	
	National Scholarship for Excellence (Sci. & Eng.) , Korea Student Aid Foundation, 2017-2021 / ca. \$20,000	
	Global Talent Attraction Program (Postponed due to Covid19), Indiana University, 2020 / ca. \$4,000	
	Semester of High Honor, 2019, 2020, 2021	
	Senior Mentoring Program Scholarship, 2019,2020	
Skills	Computer Language : Python, R, Matlab, C, C++, Cython	
	Workflow for Reproducible Data Analysis : Snakemake	
	Analysis Modules : Pytorch, Sci-kit Learn, Numpy, Scipy, Pandas, igraph, NetworkX	
	Languages : English (Proficient), Korean (Native)	
Service	Journal Reviewer	
	• 2025 Nature Matters Arising • 2025 Network Science • 2025 npj Complexity	
	Conference Reviewer	
	• International Conference on Computational Social Science 2025 • International Conference on Computational Social Science 2024 • International Society for Scientometrics and Informetrics 2023 conference • International Workshop on Cyber Social Threats (CySoc '23) • International Workshop on Cyber Social Threats (CySoc '24)	
References	Dr. Yong-Yeol Ahn , <i>Associate Professor</i>	

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Dr. Sadamor Kojaku, *Assistant Professor*
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